Load, Capacity, and Boundary: Minimal Constraint Model in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

Cara Tonucci
Independent Researcher
April 14, 2026 — Version 1
Cara.Tonucci@gmail.com
ORCID: https://orcid.org/0009-0003-9048-5821

Abstract

This document defines a minimal constraint model for the description of stability, instability, and collapse in human systems. The model is specified over three invariant variables: Load, Capacity, and Boundary (*L*, *C*, *B*). System state is determined by the relationship between these variables under conditions of time and accumulation.

The purpose is to establish a minimal and sufficient formal description of system behavior without recourse to interpretive, psychological, or domain-specific constructs. The variable set is fixed, admissible state conditions are defined, and a constrained formal anchor is provided for citation, application, and downstream elaboration.

No additional variables are introduced. No application-layer constructs are required for baseline system description. All valid downstream use inherits the constraints established here.

I. Scope

This document addresses only the minimal conditions required to determine whether a human system remains stable, enters instability, or undergoes collapse under load. It does not address meaning, interpretation, identity content, moral evaluation, or domain-specific intervention.

The model functions as a first-order anchor. It specifies the minimum invariant structure necessary to describe system behavior under demand while preserving non-collapse constraints and avoiding explanatory inflation.

II. System Definition

A system is defined as a bounded structure subject to load and possessing finite capacity. A system is exposed to demand, constrained in its ability to process that demand, and regulated by boundary conditions that govern load entry, containment, and transfer.

System behavior is determined by the interaction of Load, Capacity, and Boundary (L, C, B). These variables are treated as invariant at the minimal layer. They do not depend on awareness, interpretation, narrative framing, or institutional designation.

III. Variable Definitions

Load (L) is defined as total demand acting on a system. Load is cumulative, transferable, and time-dependent. It does not require recognition to function as a variable and is not altered by interpretation of the conditions producing it.

Capacity (C) is defined as the maximum load a system can process without loss of structural continuity. Capacity is finite, condition-dependent, and reducible under sustained overload. It is not equivalent to motivation, preference, or intent.

Boundary (B) is defined as the condition governing load entry, containment, and transfer. It regulates exposure by determining what enters a system, what remains contained within it, and what passes across system limits. Boundary is not assumed to be symmetric and may differ across inflow and outflow conditions.

IV. Minimal State Conditions

A system is stable when system load remains processable under current capacity and boundary conditions. Stability therefore refers to the preservation of structural continuity under demand.

Instability begins when load exceeds processing capacity under boundary conditions that permit continued inflow or fail to constrain accumulation. Instability is not identical to collapse. It is the condition in which capacity constraint has been violated while continuity remains present.

Collapse occurs when the violation of constraint is sustained such that the system can no longer maintain structural continuity under current conditions.

The minimal state conditions are expressed as follows:

Stability corresponds to conditions in which load does not exceed capacity under boundary conditions.

Instability occurs when load exceeds capacity under boundary conditions permitting continued inflow.

Collapse occurs when this condition is sustained over an interval t without boundary correction.

$$L \leq C \tag{1}$$

$$L > C \tag{2}$$

$$L > C \text{ over interval } t \tag{3}$$

V. Boundary Conditions

For minimal formal purposes, boundary is expressed directionally as B_in and B_out. This preserves the admissible condition that load entry and load discharge are not equivalent and need not be treated as symmetric.

Net accumulation is determined by the relation between inflow and outflow under boundary conditions, rather than by load magnitude alone. A system may therefore accumulate load independent demand magnitude.

Net load is expressed as follows:

$$L\_\mathrm{net} = L\_\mathrm{in}(B\_\mathrm{in}) - L\_\mathrm{out}(B\_\mathrm{out}) \tag{4}$$

If B_in exceeds B_out in effect, net accumulation occurs. Under these conditions, instability risk increases even without an observable change in system output.

Boundary asymmetry is structurally admissible and belongs to the minimal model rather than to downstream elaboration.

VI. Rate Condition

Load is not only a matter of magnitude but also of rate. The rate of load is expressed as follows:

$$dL/dt \tag{5}$$

Instability may arise when the rate of accumulation exceeds the system's processing rate, even if total load has not yet exceeded capacity at a single point in time.

This condition accounts for systems that remain functionally continuous while accumulating structurally significant demand faster than it can be processed.

VII. Temporal Condition

System state is a function of cumulative load over time rather than of instantaneous observation alone. For minimal formal purposes, cumulative exposure is represented as:

$$\int \text{ from } t_0 \text{ to } t_{\square}\ L(t)\, dt \tag{6}$$

This accounts for delayed instability, latent load conditions, and temporal separation between
structural cause and observable effect. A system may therefore appear stable at the level of surface output while already carrying accumulated load sufficient to produce later instability.

VIII. Generality

The Load–Capacity–Boundary model applies across individual, organizational, and distributed systems. It does not depend on domain-specific constructs, interpretive frameworks, or psychological variables. All valid applications at the minimal layer must reduce to the same invariant variable set.

IX. Constraint

All valid system descriptions under this model must reduce to Load, Capacity, and Boundary, preserve variable invariance, and avoid the introduction of additional explanatory variables at the minimal layer. Downstream applications may describe context, but they may not redefine the variable set or alter the state conditions fixed here.

X. Statement

The Load–Capacity–Boundary model provides a minimal and sufficient description of stability and collapse. No additional variables are required for baseline system description. All further elaboration remains downstream of this constraint set and inherits its limits.

XI. System Reference

The system defined in this document is specified in *Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*, DOI: 10.5281/zenodo.19075948.

All valid system descriptions reduce to Load, Capacity, and Boundary.

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

XII. Copyright